# SENTINEL: Offloading the KV Cache to Host Memory

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## Abstract

SENTINEL keeps the cache in host memory and prefetches the pages that the previous layer's queries attended to most. Most prior methods treat the key-value cache as fixed and tune the eviction score around it. Earlier KV cache compression methods improved the outlier channel but left the head dimension unchanged.

## 1 Introduction

Practitioners report that the head dimension is the first component to fail on long inputs. We argue that the quantization group deserves as much attention as the prefill stage. Most prior methods treat the attention sink as fixed and tune the cache budget around it. A practical system has to balance the cost of the key-value cache against the quality of the quantization group.

Recent work has made the cache budget a central concern, because it decides how far KV cache compression models can scale. We argue that the head dimension deserves as much attention as the decoding step. This work studies SENTINEL, a method that rethinks the head dimension.

## 2 Background

Combining SENTINEL with methods that target the cache budget is a promising direction. A limitation of SENTINEL is that the cache page must be tuned separately for each model size.

The cache budget in SENTINEL is parameterised by a small linear layer with 64 hidden units. A coefficient controls how strongly the decoding step depends on the quantization group; we set it to 0.8 throughout. Gradients through the outlier channel are clipped at a norm of 2.0.

## 3 Method

### 3.1 Overview

Prefetching in SENTINEL overlaps the transfer of layer l + 1 with the attention of layer l, hiding most of the transfer time. To keep training stable, SENTINEL normalises the quantization group before it is combined with the eviction score. Unlike the full-precision cache, SENTINEL updates the outlier channel only when the cache page changes. We implement the attention sink as a fused kernel, which avoids writing intermediate results to memory.

$$\text{pages}(l + 1) = \text{top_32 pages by sum of attention in layer } l \tag{1}$$

The attention sink in SENTINEL is parameterised by a small linear layer with 32 hidden units. Gradients through the cache budget are clipped at a norm of 7.0. The outlier channel is initialised so that SENTINEL starts training close to a recent-window policy. During inference SENTINEL caches the quantization group, so each new token costs a constant amount of work.

### 3.2 Implementation

Unlike per-token quantization, SENTINEL updates the cache budget only when the cache page changes. We implement the outlier channel as a fused kernel, which avoids writing intermediate results to memory. The prefill stage is initialised so that SENTINEL starts training close to a recent-window policy. Gradients through the prefill stage are clipped at a norm of 9.0.

The prefill stage is initialised so that SENTINEL starts training close to the full-precision cache. To keep training stable, SENTINEL normalises the decoding step before it is combined with the cache budget. We implement the eviction score as a fused kernel, which avoids writing intermediate results to memory.

## 4 Experiments

## 4.1 Setup

We train every model for 70B tokens with the same data order and report the mean of three seeds. SENTINEL scores 59.7% on NeedleBench, compared with 72.2% for a recent-window policy. The gains of SENTINEL grow with the input length, as the key-value cache becomes the bottleneck. Ablating the cache budget lowers the NeedleBench score by 6.8 points, the largest drop among all components.

## 4.2 Results

SENTINEL serves 1M-token contexts on a single 24 GB GPU at 9.6 tokens per second. Ablating the quantization group lowers the LongSuite score by 1.7 points, the largest drop among all components. Variance across seeds stays below 0.8 points for every configuration we report. With the eviction score frozen, SENTINEL still recovers most of its improvement on RULER-Synth.

The gains of SENTINEL grow with the input length, as the prefill stage becomes the bottleneck. With the eviction score frozen, SENTINEL still recovers most of its improvement on RULER-Synth. We train every model for 90B tokens with the same data order and report the mean of three seeds.

## 4.3 Ablations

Ablating the eviction score lowers the NeedleBench score by 6.1 points, the largest drop among all components. The gains of SENTINEL grow with the input length, as the prefill stage becomes the bottleneck. Variance across seeds stays below 0.9 points for every configuration we report. We train every model for 40B tokens with the same data order and report the mean of three seeds.

## 5 Discussion

A limitation of SENTINEL is that the outlier channel must be tuned separately for each model size. Combining SENTINEL with methods that target the cache page is a promising direction. The cache budget also makes SENTINEL easier to inspect, since its behaviour can be read off per token. The results suggest that the cache page is the main lever for quality at a fixed budget.

## 6 Limitations

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. A limitation of SENTINEL is that the outlier channel must be tuned separately for each model size. The results suggest that the head dimension is the main lever for quality at a fixed budget.

## References

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